Autonomous Intelligent Systems,
Institute for Computer Science VI, University of Bonn

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Assignments for module Technical Neural Networks (MA-INF 4204), WS2627

Assignment sheet 1, due: Tuesday 20.10.2026

13.10.2026

Remark: The assignments are designed to intensify the work with the research topics presented within the lecture, and to help you practice typical tasks of Technical Neural Networks. You will need more than 50% of all possible points and at least two presentations to be admitted to the exam. Please work in working groups of three or four persons.
You have to hand in the solutions by Tuesday before the start of the lecture, by uploading the solution as a .pdf file to eCampus.
You have to put the names of all (3 or 4) participating students in the solution explicitly.

Assignment 1 (2 Points)

Name and describe (briefly) the 4 important basic principles of a technical neuron that have been taken from biology.

Assignment 2 (1 Point)

Name a neural network paradigm that has NOT been mentioned within the lecture on Tuesday 13 Oct, 2026 that you would like to learn more about.
Please give the references where you have found this paradigm.

Assignment 3 (1 Point)

Name and characterize (in one sentence each) the **five** historical phases of neural network research.

Assignment 4 (2 Point)

Please explain in your own words the function and the task of the so called BIAS, and the BIAS-weights for a technical neuron.
Use formulas and a sketch/diagram **of your own** to support your explanation.

Assignment 5 (3 Points)

A simple Perzeptron with two inputs x_1, x_2, BIAS, and transferfunction $y = f(z) = sgn(z)$, separates the two dimensional input space into two parts with help of a line g.

Calculate for this Perzeptron the weights w_1, w_2, w_b in such a way, that the line g separates the given $P = 8$ patterns (${}^p x_1$, ${}^p x_2$; ${}^p \hat{y}$) into two classes.
Depict the given patterns and the resulting line g.

${}^1X = (0, 0; +1)$, ${}^2X = (2, 3; -1)$,
${}^3X = (-1 - \varepsilon, 3 - \varepsilon; +1)$, ${}^4X = (-1 + \varepsilon, 3 + \varepsilon; -1)$,
${}^5X = (-1, 2; +1)$, ${}^6X = (3, 1; -1), ;$
${}^7X = (5, -2 + \varepsilon; -1)$, ${}^8X = (5 - \varepsilon, -2; +1)$
Remark: $0 < \varepsilon \ll 1$.

Assignment 6 (3 Points)

Assignment 6 has to be operated using AI help.

Report which AI-System helped you, which prompts you have decided to use. What the percentage did the AI-system contribute to your solution? In case you had to re-set your AI-session to get an answer, report why, when and how often. Have you discarded some of the AI-generated answers? If yes, Why?

Construct the Boolean functions AND, NAND, NOR with one threshold type neuron per Boolean function. Draw them, and give the values for weights and thresholds
(it is favorable to have weights and thresholds with real valued numbers).